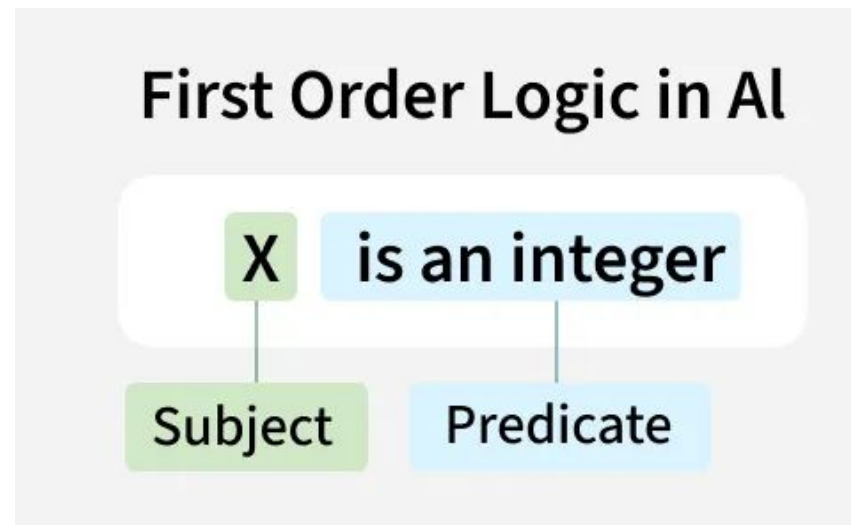


First Order Logic

First-Order Logic (FOL)

- ▶ First-Order Logic (FOL), also known as predicate logic is a ways to represent knowledge in Artificial Intelligence.
- ▶ It helps to describe both individual facts and the relationships between objects in a clear and organized manner.
- ▶ By building on propositional logic and adding quantifiers and predicates, FOL allows us to express more complex relationships and make decisions based on logical reasoning.



Key components of First-Order Logic

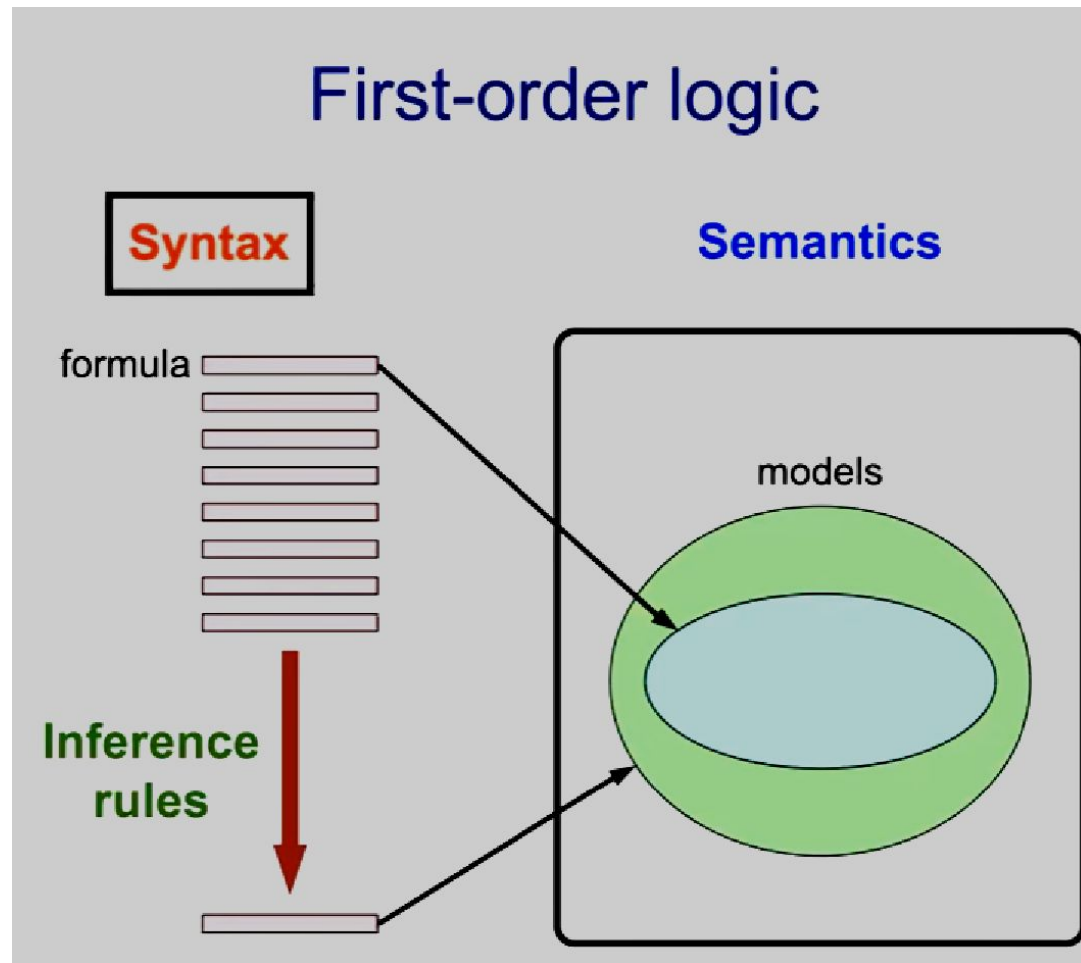
- ▶ **Constants:** represent specific objects or entities (Example: Alice, 2, NewYork).
- ▶ **Variables:** stand for unspecified objects or entities (Example: x , y , z).
- ▶ **Predicates:** define properties or relationships (Example: Likes(Alice, Bob) means "Alice likes Bob").
- ▶ **Functions:** It map objects to other objects (Example: MotherOf(x) refers to the mother of x).
- ▶ **Quantifiers:** define the scope of variables:
 - ▶ Universal Quantifier
 - ▶ Existential Quantifier
 - ▶ Logical Connectives

Key components of First-Order Logic

➤ Quantifiers:

- ▶ **Universal Quantifier** ($\forall$): Applies a predicate to all elements (Example: $\forall x (Person(x) \rightarrow Mortal(x))$ means "All persons are mortal").
- ▶ **Existential Quantifier** ($\exists$): Shows the existence of at least one element (Example: $\exists x (Person(x) \wedge Likes(x, IceCream))$ means "Someone likes ice cream").
- ▶ **Logical Connectives**: Include conjunction ($\wedge$), disjunction ($\vee$), implication ($\rightarrow$), biconditional ($\leftrightarrow$) and negation ($\neg$).

Syntax and Semantics of First-Order Logic



Syntax and Semantics of First-Order Logic

- FOL's syntax defines how to construct valid logical expressions, while semantics gives meaning to those expressions based on an interpretation which provides a domain of discourse and assigns meanings to constants, predicates and functions.
- FOL represents reasoning about knowledge in Artificial Intelligence.
- *For example, consider the domain of natural numbers. The predicate $\text{GreaterThan}(x, y)$ holds if x is greater than y . If $x = 5$ and $y = 3$, $\text{GreaterThan}(5, 3)$ is true.*
- The syntax of first-order logic consists of symbols and rules for constructing well-formed formulas (WFFs), which are statements or formulas in the language of FOL. The syntax encompasses the language constructs used to express knowledge and relationships within a domain.

Syntax and Semantics of First-Order Logic

► Predicates:

- Predicates express properties, relations, or conditions that hold between objects. They describe the state of the world or assert facts about entities within the domain. Examples include:
 - "IsHuman(x)"
 - "IsParent(x, y)"

► Quantifiers:

- Quantifiers in first-order logic allow for the specification of statements about the entirety or existence of objects within the domain.
- **Universal quantifiers ($\forall$):** Statements that hold for all objects.
- **Existential quantifiers ($\exists$):** Statements that hold for at least one object.

Syntax and Semantics of First-Order Logic

► Well-Formed Formulas (WFFs):

- Well-formed formulas (WFFs) in first-order logic (FOL) are expressions constructed according to the syntactic rules of FOL, representing meaningful statements about the world. These formulas serve as the building blocks for encoding knowledge and reasoning in AI systems.
- **Syntax Compliance:** WFFs adhere to the syntax rules of first-order logic, which define how terms, predicates, quantifiers, and logical connectives can be combined to form valid expressions.
- **Symbolic Representation:** WFFs consist of symbols representing terms (constants, variables, and functions), predicates (relations), quantifiers ($\forall$, $\exists$), and logical connectives ($\wedge$, $\vee$, $\rightarrow$, $\neg$).
- **Quantifier Scope:** WFFs maintain clear quantifier scope, ensuring that quantifiers bind variables appropriately within the formula. The scope of quantifiers affects the interpretation and meaning of the formula.
- **Complexity and Nesting:** WFFs can range from simple atomic formulas to complex nested structures involving multiple quantifiers and connectives. Proper nesting and grouping of subformulas are essential for clarity and unambiguous interpretation.

Syntax and Semantics of First-Order Logic

► *Semantics in first-order logic:*

- deals with the interpretation of sentences and formulas within the framework of a mathematical model. It provides a way to assign meanings to the symbols and structures used in first-order logic.
- The semantics of first-order logic involve defining what makes a formula true or false in a given interpretation (also called a model). An interpretation consists of:
 - **A domain of discourse:** This is the set of objects over which the variables range.
 - **Interpretations of constants:** Each constant is mapped to an element in the domain.
 - **Interpretations of predicates:** These are mappings that determine whether a predicate holds true for a particular tuple of objects from the domain.
 - **Interpretations of functions:** These mappings assign values to functions based on the values of their arguments.
- **Universal quantification ($\forall$)** asserts that a statement holds true for all objects in the domain.
- **Existential quantification ($\exists$)** asserts that there exists at least one object for which the statement is true.

Syntax and Semantics of First-Order Logic

- ▶ Alice and Bob both know arithmetic.

$$\textit{Knows}(\textit{Alice}, \textit{arithmetic}) \wedge \textit{Knows}(\textit{Bob}, \textit{arithmetic})$$

- ▶ All students know arithmetic.

$$\forall x \textit{Student}(x) \rightarrow \textit{Knows}(x, \textit{arithmetic})$$

$$\forall x (\textit{King}(x) \wedge \textit{Greedy}(x)) \rightarrow \textit{Evil}(x)$$